

Total No. of Questions : 8]

SEAT No. :

PE-2477

[Total No. of Pages : 3

[6583]-1
T.E. (Civil)
WASTE WATER ENGINEERING
(2019 Pattern) (Semester - VI) (301012)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.
- 5) Use of scientific calculators is allowed.

- Q1)** a) Explain with neat sketch working process of activated sludge process. [5]
b) Explain the term sludge volume index (SVI), hydraulic retention time (HRT), BOD Loading Rate (VLR) and F/M ratio in detail. [5]
c) Design of conventional activated sludge treatment plant for an average operating data is as follows. [8]

Wastewater flow = 32000 m³/d

Volume of aeration tank = 10500 m³

Influent BOD = 250 mg/l

Effluent BOD = 20 mg/l

Mixed liquor suspended solids = 2500 mg/l

Effluent suspended solids = 30 mg/l

Waste sludge suspended solids = 9700 mg/l

Quantity of waste sludge = 220 m³/d

Based on the information above determine,

- i) Aeration period
- ii) Food to microorganism ratio (F/M ratio)
- iii) Percentage efficiency of BOD removal
- iv) Sludge age (days)

OR

P.T.O.

Q2) a) Define sludge bulking. Explain the control measures for the sludge bulking. [5]

b) With the help of diagram explain the microbial growth pattern in batch and continuous systems. [5]

c) An average operating data for conventional activated sludge treatment plant is as follows : [8]

Sewage flow = 4000 m³/day

Volume of aeration tank = 15000 m³

influent BOD = 250 mg/lit

Effluent BOD = 20 mg/lit

Mixed liquor suspended solids = 3000 mg/lit

Effluent suspended solids = 30 mg/lit

Waste sludge suspended solids = 9700 mg/lit

Quantity of waste sludge = 220 m³/d

Determine :

i) Aeration period

ii) Food to microorganism's ratio

iii) Percentage of efficiency of BOD removal

iv) Sludge age

Q3) a) Explain with neat sketch the construction and working of oxidation pond. [5]

b) Enumerate principle, advantages and disadvantages aerated lagoon. [5]

c) Design an oxidation pond for a colony of 4000 population. The sewage flow is 120 lit/person/day BOD₅ is 300mg/l. Assume necessary required data. [7]

OR

Q4) a) Explain the construction and working of trickling filter with neat sketch. [5]

b) Determine the size of a high rate trickling filter for the following data : [9]

Sewage flow = 4.5 MLD

Recirculation ratio = 1.5

BOD of sewage = 200 mg/l

BOD removed in primary sedimentation tank = 30 %

Final effluent BOD = 20 mg/l

Depth of filter = 2 m.

c) Define the terms phytoremediation and root zone technology. [3]

- Q5) a)** Explain with neat sketch the construction and working of septic tank. **[9]**
b) With neat sketch explain anaerobic lagoon with advantages and disadvantages. **[9]**

OR

- Q6) a)** Explain working principle and applications of SBR and MBR. **[9]**
b) Design a septic tank for 350 Users. Water allowance is 150 liters per head per day. Also design a suitable soil absorption system if the percolation rate is 4 min/cm and depth of ground water table below GL is 1.5 m. **[9]**

- Q7) a)** Explain the construction and working process of anaerobic digestion tank. **[5]**
b) Write a note on gober gas plant. **[5]**
c) Explain the various methods employed for the sludge disposal along with advantages and disadvantages. **[7]**

OR

- Q8) a)** Write a short note on grey water recycling. **[5]**
b) Discuss the waste water reuse opportunities in municipal industrial and agricultural sectors. **[7]**
c) Write a short notes on sludge drying beds. **[5]**

